TEAM NUMBER: 165

M1: Approach — Gamified Financial Literacy Platform

An interactive, risk-free economic progression game for Grade 11–12 students

1. Problem Statement & Our Understanding

- **Audience:** Grade 11–12 students (ages 16–18) from underserved communities across Hyderabad, who are about to face real money decisions — stipends, UPI payments, credit offers — with little practical training.
- **Core friction:** Conventional financial education is abstract and text-heavy, causing drop-off and failing to build lasting decision-making habits.
- **Objective:** Let students learn by doing — make financial decisions, see tangible consequences, and retry without any real-world risk.
- **Coverage:** Systematically address Vanguard's four pillars — Plan & Budget, Save & Manage, Understand Investing, Stay Financially Safe.
- **Guardrails:** Strictly educational. We simulate mechanics (compounding, APR, fraud tactics) but give no investment advice, stock tips, or product endorsements.
- **Accessibility:** Trilingual (English, Telugu, Hindi), mobile-first, and light enough for budget smartphones and school lab browsers.

2. Idea & Abstract

Inspired by civilisation-rebuilding stories (e.g., Dr. Stone), the player wakes up after an economic “Great Reset” in a world where modern finance has vanished. Progress means rediscovering — by hand — why money, budgets, credit, investing and digital safety exist. All realms use one realistic currency (Rupees, ₹) so lessons transfer directly to real life.

A **fast-paced ~3-minute prologue** replaces long tutorials: barter, the double coincidence of wants, and needs vs. wants. Clearing it opens the **Dimensional Nexus** of portals:

Realm	Pillar	Simulation objective
0. Genesis Hub (Prologue)	Plan & Budget	Allocate a starter ₹2,000 across needs vs. wants, then survive a surprise emergency expense.
1. Citadel of Debt	Save & Manage	Compare moneylender vs. bank terms; feel APR compounding; separate productive from harmful debt.
2. Grand Exchange	Understand Investing	Watch compound growth over time; compare diversified index baskets vs. single risky bets.
3. Cyber Frontier	Stay Financially Safe	Defend a wallet from phishing SMS, fake UPI collect requests, spoofed QR codes, OTP and lottery scams.

- **Story Mode:** narrative progression where decisions unlock the next world.
- **Custom Mode (Skill Map):** an unlocked node-based flowchart so students and NGO workshop leads can jump straight into any simulator for targeted practice.

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3. High-Level Approach & Architecture

A responsive single-page web app (React + TypeScript + Vite + Tailwind, HTML5 Canvas for mini-games) with a lightweight backend (Node.js/Express or FastAPI, SQLite).

FRONTEND CLIENT (browser)
Narrative Engine (branching story) | Mini-Game Simulators (budget / debt / investment / fraud)
Financial Math Core (compound interest, APR, amortisation) | i18n Translation Layer (EN / TE / HI)

▼ anonymous progress & metrics

BACKEND & DATA LAYER
Anonymous Session Store (zero PII) | Progress & Mastery Tracker (Pillars 1–4 scorecard)
NGO Facilitator Telemetry (aggregate cohort completion and module scores only)

4-stage loop in every chapter: (1) Scenario & dilemma → (2) 15–30 s micro-lesson → (3) zero-risk simulation with instant visual feedback → (4) consequence & branching (wallet balance changes, next portal unlocks). Poor choices trigger a diagnostic replay explaining what went wrong, never a punishing game over.

- **Localisation:** modular JSON key-value dictionaries; language switches instantly at runtime without reloading the scene.
- **Privacy:** no names, phone numbers or emails collected; only anonymous session IDs and aggregate stats for NGO facilitators.
- **Performance:** small bundle, lazy-loaded realms and lightweight assets for slow 3G/4G connections.

4. Planned Testing Methodology

Test layer	Method	Success criterion
Financial math (unit)	Vitest/Jest tests for compound interest, APR, loan amortisation, budget surplus, Rule of 72.	Results match hand-checked reference values; no rounding drift.
Localisation coverage	Automated check that every key exists in EN, TE and HI JSON files.	100% key coverage; no missing or fallback strings.
Branching & state logic	Automated traversal of all Story Mode paths and Custom Mode nodes.	No deadlocks, broken prerequisites, or infinite loops.
Mobile UI	Viewport tests from 360×640 to 414×896 in all three languages.	No text clipping or layout overflow.
Performance	Throttled 3G network and low-end CPU profiles in browser dev tools.	Fast first load; smooth interaction on entry-level phones.
Educational consequences	Scripted playthroughs choosing bad options (UPI scam, high-APR lender).	Each triggers a clear explanation and retry, not a dead end.

We will also run a short informal usability check with a few peers (time-to-complete, points of confusion) and iterate before the final submission.